Customer Activity Analytics:
Build a web application for a financial-services operator.

Goal:

The idea of this assignment is to develop a tool used by customer care operators to analyze the activities and risk of some clients.
It should be presented as a dashboard where the operator can see and monitor all the information required.
Part of the assignement is to interact with LLMs to providing analysis on de date (see below the spec).
The candidate should generate this application using AI tools and agents. Part of the assessment will be around how this application was built and which metholodogy was used to achieve the result.

Specification:

1. The operator should be able to search for a customer by Customer ID and review the customer's activity. Customer activity is stored in a relational database and consists of three types: *card* activity, *payment* activity, and *cryptocurrency* activity

2. The application should provide a clear overview of the customer's activity and allow the operator to request an AI-powered analysis of that activity.
The AI analysis should provide a risk level, summarize the findings, and provide recommendations.

3. Add login possibility by different operators

4. RAG should be used for retrieving relevant unstructured knowledge and policies.

5. AI analysis results should be persisted and available for later review.

Technologies:

Preferred technologies: Java 17+, Spring Boot, Hibernate/JPA, React, relational database. Candidate may choose supporting technologies and the LLM provider (or use of stubs for the actual calls to LLM) .

Tables:

transactions

transaction_id	UUID (PK)	Unique transaction identifier
customer_id	UUID (FK → customers)	Owner of the activity
activity_type	ENUM	'CARD', 'PAYMENT', 'CRYPTO'
amount	DECIMAL(18,2)	Transaction amount
currency	VARCHAR(10)	ISO currency code or crypto ticker
status	VARCHAR	Completed / Pending / Failed / Reversed
created_at	TIMESTAMP	When the transaction occurred

card_activity

transaction_id	UUID (PK, FK → transactions)	Links to base transaction
card_pan	VARCHAR	Masked PAN (e.g. ****1234)
card_type	VARCHAR	Debit / Credit / Prepaid
merchant_name	VARCHAR	Merchant descriptor
mcc_code	VARCHAR(4)	Merchant category code
card_present	BOOLEAN	Physical vs card-not-present
authorization_code	VARCHAR	Issuer auth code
decline_reason	VARCHAR	Reason if declined (nullable)

payment_activity

transaction_id	UUID (PK, FK → transactions)	Links to base transaction
payment_method	VARCHAR	ACH / Wire / SWIFT / P2P
sender_account	VARCHAR	Originating account IBAN
receiver_account	VARCHAR	Destination account IBAN
receiver_bank_country	CHAR(2)	Beneficiary bank country

crypto_activity

transaction_id	UUID (PK, FK → transactions)	Links to base transaction
blockchain	VARCHAR	BTC / ETH / etc.
wallet_address_from	VARCHAR	Source wallet
wallet_address_to	VARCHAR	Destination wallet
tx_hash	VARCHAR	On-chain transaction hash
exchange_name	VARCHAR	exchange, if any

Risk layer

risk_assessments (supports multiple risk signals per transaction over time)

assessment_id	UUID (PK)	Unique assessment record
transaction_id	UUID (FK → transactions)	Transaction being scored
rule_id	UUID (FK → risk_rules)	Rule that triggered
triggered_at	TIMESTAMP	When the rule fired
score_contribution	DECIMAL(5,2)	Points added to risk_score

risk_rules

rule_id	UUID (PK)	Rule identifier
rule_name	VARCHAR	e.g. "High-value cross-border payment"
applies_to	ENUM	CARD / PAYMENT / CRYPTO / ALL
threshold_logic	TEXT	Rule condition
weight	DECIMAL(5,2)	Default score weight

How to provide results:

Please provide the project as a Git repository together with a README describing how to run the application, the architecture, the main design decisions, and any assumptions.

Expected delivery: Git repository (share for recruitment) + short demo (10–15 minutes).

Database schema in attachment.

Extras:
Provide summary of LLMs of choice and short summary of agent instructions given.